Stock Ordering Process Improvement — Business Analysis Case Study

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Executive summary

Stock ordering at a 6-aisle IGA supermarket is reactive. Items are reordered only once both the shelf and the top capping are empty, and quantities follow Metcash deals rather than actual sales.

This takes about 6 staff-hours a week (estimate) and causes two opposite failures. Items on special are under-ordered and sell out, while other lines, especially bread, are over-ordered and expire.

Recommendation: move to data-informed ordering using tools the store already has. This means par levels built from POS sales history, the Metcash promotions calendar to size special orders, labelled back-storage locations, a customer out-of-stock log, and a rotating stock lead on each order day. It needs no new software and can be tested in an 8-week pilot.

Background and problem statement

The store is an IGA supermarket in Melbourne, supplied by Metcash. The analyst works there as a team member, filling stock and serving on the register, and is not responsible for ordering.

Store fact	Detail
Size	6 aisles, plus back storage and a cool room
Opening hours	6 am to 11 pm
Staff	12 in total, 3 per shift
Ordering	3 days a week, about 2 hours each (estimate), led by the manager with one team member
Deliveries	Tuesday, Wednesday and Friday

Store fact	Detail
Tools	Handheld scanner, Metcash supplier portal, Microsoft-based POS

Problem statement: The manager and ordering staff decide what to order by walking the aisles and checking back storage item by item, and decide how much to order from Metcash discounts and specials. Because no sales data is used, specials regularly sell out before the next delivery, while slow lines and short-life stock such as bread expire. Staff lose time on each order day, and customers asking when items like rice or yogurt will return get no clear answer.

Stakeholder analysis

Two team members were interviewed informally. Both point to the same root cause: order quantities are guessed, not based on what sells.

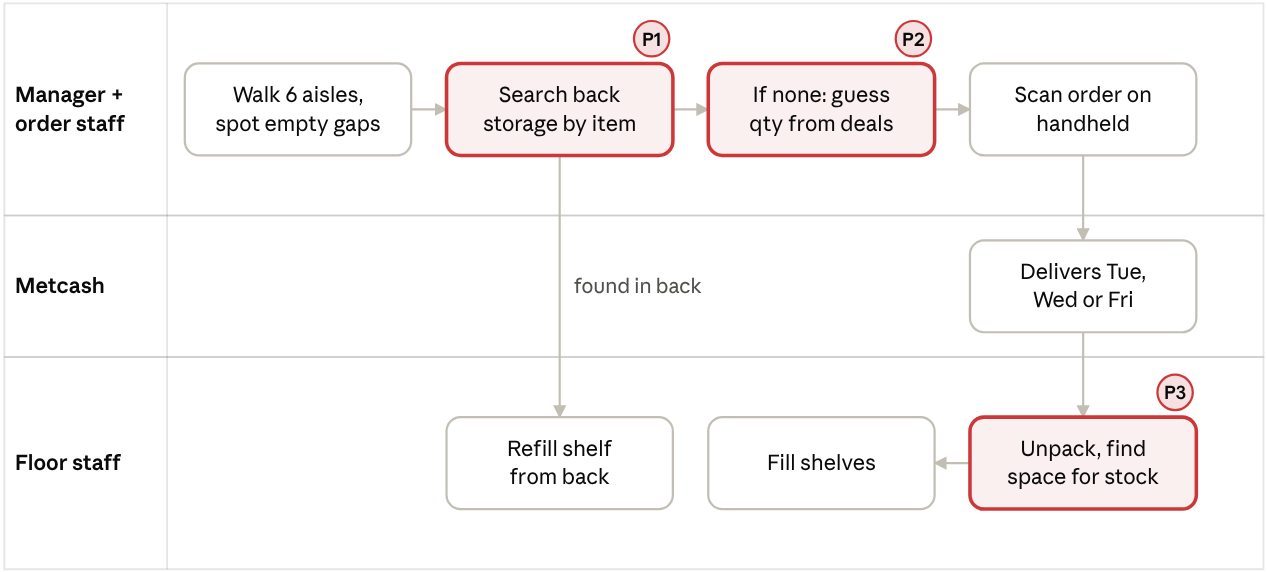
Stakeholder	Role in the process	How engaged	Pain points	What they need
Team member (coworker)	Fills stock, helps with ordering, unpacks deliveries	Interview	Checking back storage for every product is slow. Bulk special stock has no planned space. Over-ordered stock, mostly bread, expires before it sells.	A dedicated person watching stock levels, expiry and ordering

Stakeholder	Role in the process	How engaged	Pain points	What they need
Team member (analyst)	Fills stock, serves on the register	Own observation	Specials often run out several times a month. Customers ask when out-of-stock items such as rice and yogurt will return, and staff can't say.	A way to see and share expected restock dates
Store manager	Leads ordering, approves all orders	Not yet interviewed	To be confirmed	To be confirmed
Metcash	Supplier, sets deals and delivery days	Indirect	Deals and specials drive order sizes	Orders placed in time for delivery days
Customers	Buy the stock	Indirect, through requests at the register	Items on special and staples unavailable	Products on the shelf, or a clear restock date

As-is process and pain points

Stock is reordered only after it has run out

As-is ordering process on each order day (3 a week) · pain points P1 to P3 in red



As-is ordering process · 3 swimlanes, 3 pain points

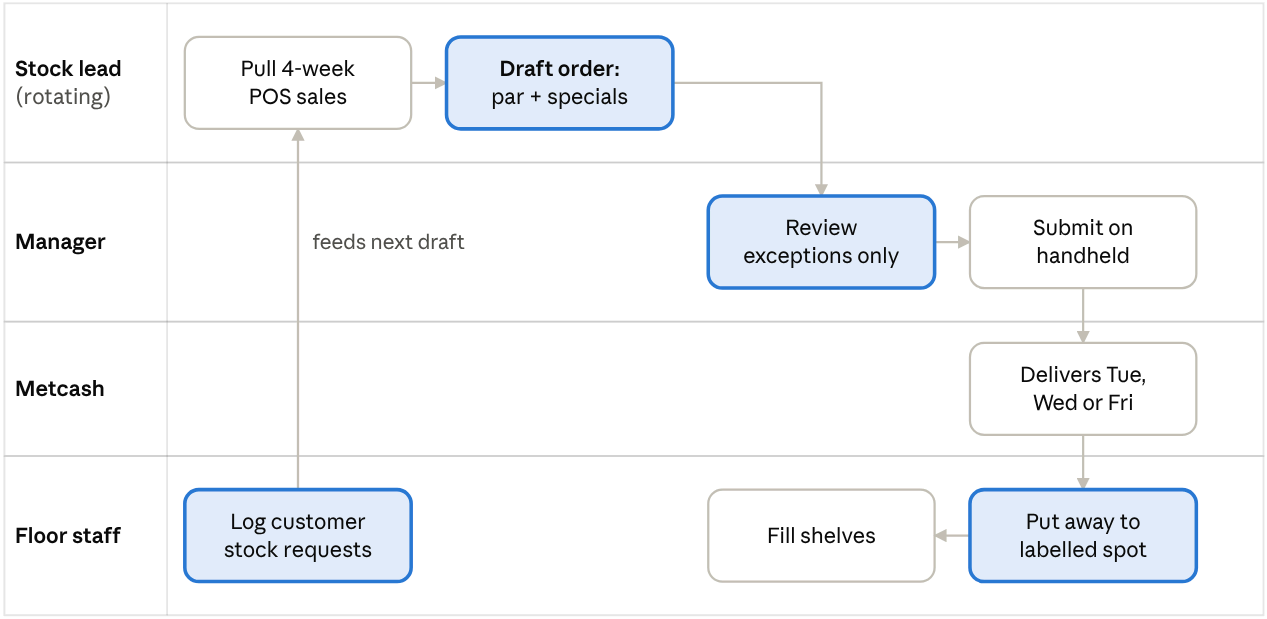
Ordering starts only when a shelf and its top capping are both empty, so every order reacts to a stock-out that has already happened.

#	Pain point	Effect	Raised by
P1	Back storage checked item by item before ordering	Slow: part of about 2 hrs per order day (estimate)	Coworker
P2	Quantities set by Metcash deals and specials, not sales	Specials under-ordered and sell out, often several times a month; other lines over-ordered	Coworker, analyst
P3	No plan for where bulk special stock will go	Time lost finding space on delivery days	Coworker
P4	Over-ordered short-life stock expires, mostly bread	Waste; bread returned to the supplier for credit	Coworker
P5	No record of out-of-stock items or restock dates	Staff can't tell customers when items like rice or yogurt will return	Analyst

To-be process

Orders are drafted from sales data before shelves run empty

To-be ordering process · new or changed steps highlighted



To-be ordering process · 4 swimlanes, 4 new or changed steps

A rotating stock lead drafts each order from sales data before 9 am, so the manager only reviews exceptions instead of walking every aisle.

Change	Fixes
Draft order from 4-week POS sales and par levels	P2: quantities based on what sells
Size specials using the Metcash promotions calendar and past uplift	P2: fewer specials selling out
Labelled back-storage bays, with space planned for large special orders	P1 and P3: faster checks and put-away
Weekly waste list (reductions, write-offs, bread credits)	P4: over-ordered lines cut back
Customer out-of-stock log at the register, with next delivery day shown	P5: demand recorded and customers told when stock returns

Requirements and user stories

Five of the ten stories are Must-haves, covering order quantities, specials, back storage and waste. Priorities use MoSCoW (Must, Should, Could, Won't).

ID	User story	Acceptance criteria	Priority
US1	As the store manager, I want a suggested order quantity for each item based on the last 4 weeks of sales, so that orders are not guesses.	Given POS sales are exported weekly, when the stock lead drafts an order, then each item shows average weekly sales, its par level and a suggested quantity.	Must
US2	As the store manager, I want upcoming Metcash specials listed with the sales uplift from the last time each was on special, so that I order enough to last until the next delivery.	Given the Metcash promotions calendar, when a special starts within 7 days, then the draft order shows the item, the last uplift and a suggested quantity.	Must
US3	As a team member doing the order, I want to know which items are in back storage and where, so that I don't search item by item.	Given back storage has labelled bays, when I check an item, then I can find it or confirm it is absent in under 1 minute.	Must
US4	As the store manager, I want a weekly list of items reduced, written off or returned for credit, so that I can cut over-ordered lines.	Given reductions and bread credits are logged, when the week closes, then I see the top 10 lines by units wasted.	Must
US5	As the stock lead, I want an order-day checklist, so that any trained team member can prepare the draft order.	Given the checklist, when a newly trained team member prepares an order, then the manager approves it with only exception changes.	Must
US6	As a register team member, I want to log a customer request for an out-of-stock item in under 30 seconds, so that the manager sees missed demand.	Given a shared log at the register, when a customer asks for a missing item, then I record the item and date in under 30 seconds.	Should

ID	User story	Acceptance criteria	Priority
US7	As a register team member, I want to see the expected restock date of an out-of-stock item, so that I can tell the customer.	Given an item is on order, when a customer asks, then I can see its next delivery day.	Should
US8	As a floor team member, I want to know before a large special delivery arrives where it will be stored, so that unpacking is quick.	Given a special order above normal volume, when it is placed, then a storage location is assigned before delivery day.	Should
US9	As the opening team member, I want items expiring the next day listed by aisle, so that 6 am reductions are faster.	Given expiry dates are recorded at put-away, when I open the store, then I see items expiring tomorrow sorted by aisle.	Could
US10	As the store manager, I want orders sent to Metcash automatically without review.	Out of scope for this phase. Human review stays.	Won't

Non-functional requirements

- Works on the store's existing handheld scanner, POS and a shared spreadsheet, with no new paid software during the pilot.
- Training takes no more than 30 minutes per team member.
- Draft order is ready by 9 am on each order day.
- Sales data in the draft is no more than 7 days old.

Options analysis and recommendation

Option C is recommended. It fixes the root cause, guessed quantities, at the lowest cost and risk.

Option	What it involves	Cost (relative)	Benefits	Risks
A. Do nothing	Keep walking aisles and ordering on deals	None	No change effort	Stock-outs on specials and bread waste continue; about 6 hrs a week stays spent (estimate)
B. Dedicated stock controller	Hire or assign one person to own stock, expiry and ordering (coworker's suggestion)	High: extra paid hours	Clear ownership, consistent checks	Ongoing wage cost; still guesswork without sales data; single point of failure
C. Data-informed ordering (recommended)	Par levels from POS sales, promo calendar for specials, labelled back storage, out-of-stock log, rotating stock lead on order days	Low: staff time to set up, existing tools	Targets the root cause; any trained team member can draft orders; builds a record of demand	Depends on clean POS exports; staff must keep logs up to date
D. Inventory management software	Third-party system with automatic replenishment	Medium to high: licence and setup	Most automated	Cost unknown; integration with Metcash and POS unproven; long setup

Option C also delivers most of what Option B asks for. The rotating stock lead gives ordering a clear owner each order day without adding paid hours. If the 8-week pilot succeeds, Option D can be reviewed later with real data to judge it against.

KPIs and next steps

Four KPIs will show whether the pilot worked. Baselines are measured in the first 2 weeks, and the targets below are proposals to agree with the manager.

KPI	How it is measured	Baseline	Proposed pilot target
Ordering time	Staff-hours spent on order days, logged each week	About 6 hrs a week (estimate)	30% lower
Specials stock-outs	Items on special that sell out before the next delivery, per month	Measured in weeks 1 to 2	Halved
Bread and short-life waste	Units reduced, written off or returned for credit, per week	Measured in weeks 1 to 2	25% lower
Unmet customer requests	Out-of-stock requests logged at the register, per week	Measured in weeks 1 to 2	Downward trend over the pilot

Next steps

- [ ] Present this case study to the store manager and interview them
- [ ] Log the 4 KPI baselines for 2 weeks
- [ ] Export 4 weeks of POS sales and build a par-level spreadsheet
- [ ] Label back-storage bays and record locations
- [ ] Set up the out-of-stock log at the register
- [ ] Run the 8-week pilot with a rotating stock lead
- [ ] Review KPIs with the manager and decide whether to keep, adjust or scale up

Assumptions and limitations

- All time figures are staff estimates, not measured values.
- Findings come from 2 informal interviews and the analyst's own observation. The store manager has not yet been interviewed.
- No sales, stock or waste data was used, so all targets are proposals until baselines are measured.

- The POS is assumed to export item-level sales. This needs to be confirmed.
- No staff names, sales figures or supplier prices are included.